AI-1 Spring 26 Project Proposal

Team Members

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Idea Short Name (Animal Stacker)

Idea Description

the game uses animals of different sizes (Chick 🐤, Cat 🐱, Fox 🦊, Panda 🐼, Elephant 🐘) that must be stacked correctly according to size constraints. The goal is to move all animals from the starting peg (Peg 0) to the goal peg (Peg 2) following the rule: *a larger animal cannot be placed on top of a smaller animal by using [BFS, DFS, A*] algorithms.

Agent Architecture

☐ Simple Reflex Agent

√ Model-Based Reflex Agent

√ Goal-Based Agent

√ Utility-Based Agent

Problem-Solving Framework

√ Classical Search

☐ Local Search (Optimization)

☐ Adversarial Search (Two-Player Games)

☐ Constraint Satisfaction Problem

Proposed Algorithm(s)

BFS **DFS** **A***

PEAS

Performance Measures	• Minimize the number of moves to reach the optimal solution. • Minimize the number of expanded nodes. • Minimize the execution time. • Successfully solve the puzzle without violating the size rule (never place a larger animal on a smaller one).
Environment	the environment consisting of three pegs and multiple disks of different sizes arranged on the pegs.
Actuators	• Move an animal from one pole to another. • Display the movement with smooth animation. • Show the complete solution path. • Display statistics (number of moves, expanded nodes, execution time, path cost).
Sensors	• Current state of the three poles (which animals are on each pole and their order). • Position of each animal. • Selected difficulty level and number of animals.

Environment Properties

√ Fully Observable ☐ Partially Observable	√ Deterministic ☐ Stochastic ☐ Strategic	√ Discrete ☐ Continuous

√ Single-Agent ☐ Multi-Agent	√ Sequential ☐ Episodic	√ Static ☐ Dynamic